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Reducing fractions to a common denominator

Building fractions up instead of cutting them down — the fundamental property read from left to right.

LESSON 9–11 | 3 × 40 MIN | ALGEBRA 8, §3 | STAGE 9 · 2.5

LESSON CLOCK

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Factorise denominators and build the lowest common denominator from the factors.
- Find the additional factor for each fraction.
- Rewrite two or three fractions with the same denominator.
- Recognise the reversed-bracket case $a - b$ and $b - a$.

TEXTBOOK

National	§3, pp. 18–21
Cambridge	Learner’s Book pp. 40–44
Workbook	Workbook 2.5

01 Explanation

Why a common denominator at all

You cannot add $\frac{1}{2}$ and $\frac{1}{3}$ directly because the pieces are different sizes. Cut both into sixths and the addition becomes counting. Algebraic fractions behave exactly the same way — only the “sixths” are now expressions.

The lowest common denominator

METHOD

1. Factorise every denominator completely.
2. Take **every factor that appears anywhere**, each to its **highest** power.
3. The product of these is the lowest common denominator (LCD).
4. Divide the LCD by each denominator — that quotient is the fraction's **additional factor**.
5. Multiply the numerator and denominator of each fraction by its additional factor.

Numbers work the same way. For $\frac{1}{4x}$ and $\frac{1}{6y}$: the numerical LCM of 4 and 6 is 12, the letters give xy, so the LCD is $12xy$.

A worked pattern

Bring $\frac{3}{x^2 - 9}$ and $\frac{1}{x - 3}$ to a common denominator.

$$x^2 - 9 = (x - 3)(x + 3),\ x - 3 = (x - 3)$$

Every factor that appears: $(x - 3)$ and $(x + 3)$, each to the first power. So

$$LCD = (x - 3)(x + 3)$$

The first fraction already has it — additional factor 1. The second needs $(x + 3)$:

$$\frac{3}{(x - 3)(x + 3)} \quad \text{and} \quad \frac{x + 3}{(x - 3)(x + 3)}$$

DO NOT MULTIPLY BLINDLY

Multiplying the denominators together always *works*, but it can give $(x - 3)(x + 3)(x - 3)$ where $(x - 3)(x + 3)$ was enough — and then you are cancelling it back down at the end. Factorise first, and the lowest denominator appears by itself.

When one denominator is the reverse of the other

$x - y$ and $y - x$ are not different denominators — they differ by a sign. Multiply the second fraction's numerator and denominator by -1:

$$\frac{y}{y - x} = \frac{-y}{x - y}$$

Now both fractions share the denominator $x - y$ and no extra factors are needed.

02 Worked examples

EXAMPLE 1 ***Bring $\frac{1}{x^2 - 4}$ and $\frac{1}{x^2 - 4x + 4}$ to a common denominator***

1. $x^2 - 4 = (x - 2)(x + 2)$

   First denominator.

2. $x^2 - 4x + 4 = (x - 2)^2$

   Second denominator — note the square.

3. $LCD = (x - 2)^2(x + 2)$

   (x − 2) appears to the power 2, so take it squared.

4. *additional factors:* $(x - 2)$ *and* $(x + 2)$

   LCD ÷ each denominator.

5. $\frac{x - 2}{(x - 2)^2(x + 2)}$ and $\frac{x + 2}{(x - 2)^2(x + 2)}$

   Multiply top and bottom by them.

ANSWER

$LCD = (x - 2)^2(x + 2), \; x \neq \pm 2$

EXAMPLE 2 ***Find the LCD of*** $\frac{2}{3a^2b}$ ***and*** $\frac{5}{6ab^3}$

1. $LCM(3, 6) = 6$

   The numbers first.

2. *a: highest power* a^2

   a^2 beats a.

3. *b: highest power* b^3

   b^3 beats b.

4. $LCD = 6a^2b^3$

   Multiply what you collected.

ANSWER

$6a^2b^3$

03 Interactive model

Ask the class for the LCD before you open the table — then check their answer against it.

Building the common denominator

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
$x^2 - 1$	$(x - 1)(x + 1)$	1
$x + 1$	$(x + 1)$	$(x - 1)$

LOWEST COMMON DENOMINATOR

$$(x - 1)(x + 1) = x^2 - 1$$

The first denominator already *is* the LCD — only the second has to be built up.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Common denominator	Umumiy maxraj	Общий знаменатель
Lowest common denominator	Eng kichik umumiy maxraj	Наименьший общий знаменатель
Additional factor	Qo‘shimcha ko‘paytuvchi	Дополнительный множитель
Lowest common multiple	EKUK	НОК
Highest power	Eng yuqori daraja	Наивысшая степень
Bring to a common denominator	Umumiy maxrajga keltirish	Привести к общему знаменателю
Factorised form	Ko‘paytma ko‘rinishi	Разложенный вид
Reversed bracket	Teskari qavs	Перевёрнутая скобка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The LCD of $\frac{1}{x}$ and $\frac{1}{x^2}$ is:

- x
- x^2
- x^3
- $2x$

The LCD of $\frac{1}{x-1}$ and $\frac{1}{x+1}$ is:

- $x^2 - 1$
- $x^2 + 1$
- x
- $(x - 1)^2$

To write $\frac{3}{x}$ with denominator x^2, the numerator becomes:

- 3
- $3x$
- $3x^2$
- x

$\frac{y}{y - x}$ written with denominator $x - y$ is:

- $\frac{y}{x - y}$
- $\frac{-y}{x - y}$
- $\frac{x}{x - y}$
- impossible

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *LCD of* $\frac{1}{2}$ *and* $\frac{1}{3}$

ANSWER 6

2. *LCD of* $\frac{1}{x}$ *and* $\frac{1}{y}$

ANSWER xy

3. *LCD of* $\frac{1}{2a}$ *and* $\frac{1}{3a}$

ANSWER $6a$

4. *LCD of* $\frac{1}{x}$ *and* $\frac{1}{x^2}$

ANSWER x^2

5. *LCD of* $\frac{1}{x-1}$ *and* $\frac{1}{x+1}$

ANSWER $(x-1)(x+1)$

6. *Write* $\frac{3}{x}$ *with denominator* x^2

ANSWER $\frac{3x}{x^2}$

7. *Write* $\frac{2}{a}$ *with denominator* ab

ANSWER $\frac{2b}{ab}$

Medium · the standard question

7 PROBLEMS

1. *LCD of* $\frac{1}{x^2 - 1}$ *and* $\frac{1}{x + 1}$

   ANSWER $(x - 1)(x + 1)$

2. *LCD of* $\frac{1}{2x}$ *and* $\frac{1}{3x^2}$

   ANSWER $6x^2$

3. *Bring* $\frac{1}{x - 2}$ *and* $\frac{1}{x + 2}$ *to a common denominator*

   ANSWER $\frac{x + 2}{x^2 - 4}$ and $\frac{x - 2}{x^2 - 4}$

4. *LCD of* $\frac{1}{x^2 + 2x}$ *and* $\frac{1}{x + 2}$

   ANSWER $x(x + 2)$

5. *LCD of* $\frac{1}{a^2 - b^2}$ *and* $\frac{1}{a - b}$

   ANSWER $(a - b)(a + b)$

6. *Bring* $\frac{3}{x^2 - 9}$ *and* $\frac{1}{x - 3}$ *to a common denominator*

   ANSWER $\frac{3}{x^2 - 9}$ and $\frac{x + 3}{x^2 - 9}$

7. *LCD of* $\frac{1}{4x}$ *and* $\frac{1}{6y}$

   ANSWER $12xy$

Hard · stretch and reason

7 PROBLEMS

1. *LCD of* $\dfrac{1}{x^2 - 4}$ *and* $\dfrac{1}{x^2 - 4x + 4}$

   ANSWER $(x - 2)^2(x + 2)$

2. *LCD of* $\dfrac{1}{x^2 + x}$ *and* $\dfrac{1}{x^2 - x}$

   ANSWER $x(x - 1)(x + 1)$

3. *LCD of* $\dfrac{1}{x^2 + 5x + 6}$ *and* $\dfrac{1}{x^2 + 4x + 4}$

   ANSWER $(x + 2)^2(x + 3)$

4. *Bring* $\dfrac{1}{x^2 - 1}$ *and* $\dfrac{1}{x^2 + 2x + 1}$ *to a common denominator*

   ANSWER LCD $(x - 1)(x + 1)^2$: $\dfrac{x + 1}{(x - 1)(x + 1)^2}$ and $\dfrac{x - 1}{(x - 1)(x + 1)^2}$

5. *LCD of* $\dfrac{2}{3a^2b}$ *and* $\dfrac{5}{6ab^3}$

   ANSWER $6a^2b^3$

6. *Bring* $\dfrac{x}{x - y}$ *and* $\dfrac{y}{y - x}$ *to a common denominator*

   ANSWER $\dfrac{x}{x - y}$ and $\dfrac{-y}{x - y}$

7. *LCD of* $\dfrac{1}{x^2 - 4x}$ *and* $\dfrac{1}{x - 4}$

   ANSWER $x(x - 4)$

07 Homework

Homework — 5 problems

Algebra 8, §3, pp. 18–21. Factorise every denominator before you write the LCD.

1. *Find the LCD of* $\frac{1}{x^2 - 25}$ *and* $\frac{1}{x + 5}$
2. *Find the LCD of* $\frac{1}{6ab}$ *and* $\frac{1}{9b^2}$
3. *Bring* $\frac{2}{x - 3}$ *and* $\frac{5}{x + 3}$ *to a common denominator*
4. *Bring* $\frac{1}{x^2 + x}$ *and* $\frac{1}{x}$ *to a common denominator*
5. *Write* $\frac{a}{a - b}$ *and* $\frac{b}{b - a}$ *with the same denominator*